INCH-POUND

MIL-B-43611B
31 December 1990
SUPERSEDING
MIL-B-43611A
3 September 1974

MILITARY SPECIFICATION

BAGS, PAPER, WAXED (FIELD BREAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers waxed paper bags for wrapping field bread.

1.2 Classification. Waxed paper bags shall be of the following types and minimum sizes as specified (see 6.2).

Type I - Self opening (automatic), 13 by 6 by 33 inches (Face, width, height).

Type II - Square, 13 by 6 by 35 inches.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8105

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2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

UU-B-36	-	Bags, Paper (Grocers)
A-A-203	-	Paper, Kraft, Wrapping
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Part 121)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001).

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102-1187).

Technical Association of the Pulp and Paper Industry (TAPPI)

T-401 - Fiber Analysis of Paper and Paperboard
T-402 - Conditioning Paper and Paperboard for Testing
T-403 - Bursting Strength of Paper
T-404 - Tensile Breaking Strength of Paper and Paperboard
T-405 - Wax in Papers
T-410 - Basis Weight of Paper and Paperboard
T-414 - Internal Tearing Resistance of Paper
T-477 - Blocking Resistance of Paper and Flexible Materials
T-483 - Odor of Packaging Materials
T-630 - Melting Point (Plateau) of Petroleum Wax

(Copies may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30346.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subject to first article inspection (see 6.3), in accordance with 4.3.

3.2 Materials. The materials used in the fabrication of waxed paper bags shall conform to the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Part 121). It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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3.2.1 Waxed kraft paper. The paper used in the production of waxed paper bags shall be not less than 90 percent unbleached kraft fibers, and not more than 10 percent of other chemical fibers providing the paper meets the physical requirements of table I. The paper shall be waxed both sides. Physical requirement tests shall be as specified in 4.4.2.

TABLE I. Minimum physical requirements of waxed paper

Characteristics	Requirements - minimum
Wax in paper	12 pounds per ream 1/
Basis weight of unwaxed paper	50 pounds per ream 1/, minus 5 percent, unlimited plus tolerance
Bursting strength (before waxing)	31 points per square inch
Tensile strength	17 pounds per inch of width
Tearing resistance, machine direction (before waxing)	92 grams
Blocking resistance	Slight blocking
Odor of packaging material	No evidence of foreign odor
Melting point of wax	140°F.

1/ Ream (24 by 36 inches - 500 sheets).

3.2.2 Adhesive. The adhesive used for the seams of waxed paper bags shall be waterproof to the extent that there shall be not more than 25 percent adhesive failure of the bonded area when tested as specified in 4.4.5. Seams shall be continuously bonded.

3.3 Construction. Waxed paper bags shall be constructed with an open mouth in accordance with the applicable figures in UU-B-36. The bags shall be fabricated with a thumb notch or lip in accordance with commercial practice.

3.4 Dimensions. Dimensions specified in 1.2 shall be minimum sack dimensions when measured as specified in applicable figures in UU-B-36. The edges of unopened bags shall be uniform and parallel.

3.5 Marking. Each bag shall be printed or stenciled, across one face of the bag in letters not less than 1/2 inch high, with the words "FIELD BREAD BAG". The ink used in printing or stenciling shall be commercially waterfast and sunfast and shall have no objectionable odor. The position of the printing shall be not less than 10 inches from the open top of the bag and approximately equidistant from each edge of the bag.

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3.6 Workmanship. Bags shall be uniformly constructed and shall be free from holes, tears, or other defects which would impair their usefulness. Bags shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimesions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4, and tested as specified in 4.4.5.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Components and material testing. Testing of materials described in this specification shall be as indicated in table II. The lot for the purpose of determining sample size shall be expressed in units of rolls of the waxed kraft paper. The sample shall be one piece the full width of roll by 36 inches long. Not more than one sample unit shall be drawn from any one roll and shall not be taken from the two outer coils of the roll. The inspection level shall be S-1, except that minimum sample size shall be three sample units. Results for the characteristics shown in table II shall apply to lot average. The lot shall be subject to rejection if any of the average minimum requirements of the waxed kraft paper are not met.

4.4.2.1 Conditioning of the test unit. Before testing for the characteristics in table II, all test units shall be conditioned to equilibrium in accordance with TAPPI Method T-402. Conditioning shall also apply on obtaining the basis weight of kraft paper after the wax extraction cycle.

TABLE II. Component and material tests

Characteristic	Requirement paragraph	TAPPI test method 1/	Determinations per sample unit	Report to nearest
Fiber analysis of kraft paper	3.2.1	T-401		1 percent
Weight of wax in paper	3.2.1	T-405	2	0.1 pound
Basis weight of paper after wax extraction	3.2.1	T-410	2	0.1 pound
Bursting strength	3.2.1	T-403	Average of 10-5 on each side	0.1 point
Tensile strength	3.2.1	T-404	Average of 5	0.1 pound
Tearing resistance (machine direction)	3.2.1	T-414	Average of 5	1 gram

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TABLE II. Component and material tests (cont'd)

Characteristic	Requirement paragraph	TAPPI test method 1/	Determinations per sample unit	Report to nearest
Blocking resistance	3.2.1	T-477	1	rating
Odor (foreign)	3.2.1	T-483	1	none
Melting point of wax	3.2.1	T-630	1	1°F

1/ Test procedures and controls which differ from those TAPPI methods herein may be used by the contractor if they provide a quality assurance equivalent to that specified. If the government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor shall use the test procedures set forth herein. In case of dispute as to the test results, the test methods herein shall govern.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of bags. The sample unit shall be one bag. The inspection level shall be S-4, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. End item visual defects

Examine	Defect
Construction	Bag not type specified
	Seams not continuously bonded
	Edges of unopened bag not uniform and parallel
Marking	Not as specified
	Letters less than 1/2 inch high (see 3.5)
	Not legible
	Position of printing less than 10 inches from top of bag, not equidistant from bag edges
Workmanship	Tears or holes
	Surface not clean

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4.4.4 End item dimensional examination. The paper bags shall be examined for the dimensions specified in 3.4. Only those dimensions that can be evaluated without damaging or disassembling the paper bags shall be examined. Any dimension that is less than the size specified in 3.4 shall be classified as a defect. The lot size shall be expressed in units of bags. The sample unit shall be one bag. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table IV. The lot size shall be expressed in units of bags. The sample unit shall be one bag. The inspection level shall be S-2. Any test failure shall be cause for rejection of the lot.

TABLE IV. End item testing

Characteristics	Requirement paragraph	Test method	Determinations per sample unit	Results reported as pass or fail 2/
Waterproofness of adhesive 1/	3.2.2	4.5.1	2 (one on side and bottom seam specimen)	X

1/ Failure of any specimen to pass the test shall be cause for rejection of the lot.

2/ If failure is indicated, report description thereof.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling

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Examine	Defect
Workmanship (cont'd)	Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Testing of the end item for water resistance of adhesive (see 3.2.2). Cut strips 1-inch wide across the longitudinal and bottom seams, adjusting the length so that both sheetings will extend approximately 1-1/2 inches beyond each edge of the bonded joint. Immerse the specimens in approximately 1-1/2 inches of distilled water at room temperature for not less than 24 hours; after the bath cycle, remove one specimen, blot off excess water with blotting paper, and test before removing the next strip. To test, grasp the projecting ends of the specimen with one thickness of paper between the thumb and forfinger of each hand, the forfingers resting against the bonded joint, one on each side. Separate the joints by gently pulling the ends apart with a rolling motion from the top downward with second joints of the second and third fingers successively resting against each other to act as fulcrums. Examine the manner of separation of the joint, noting the percent of separation that occurs without fiber failure of the prebonded area of each separated sheet. Failure of more than 10 percent of the specimens tested shall be cause for lot rejection.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Two hundred and fifty bags of one type only shall be uniformly stacked and assembled into a package. The package shall then be completely wrapped in the same material as specified for the bag (see 3.2.1) or in kraft paper having a minimum basis weight of 60 pounds per ream (24 by 36 inches-500 sheets) in accordance with A-A-203. The paper wrap shall overlap itself widthwise and lengthwise a minimum of four inches and shall be secured with crossties of twine. The twine shall have a minimum breaking strength of 200 pounds and the two intersections of the cord shall be interlocked. Alternatively, the bags may be stacked in quantities of 50 and either crosstied with not less than 50-pound breaking strength cord or twine or secured with a kraft paper band or sleeve having a minimum basis weight of 30 pounds per ream (25 by 36 inches-500 sheets). Two hundred and fifty bags shall then be packaged in a snug-fitting bag made from the material specified for the 250 bag wrap above. Closure of the bag shall be effected by double folding over the mouth a minimum of 1/2 inch and either gluing, stapling with not less than three staples, or taping with tape conforming to PPP-T-76.

5.1.2 Commercial preservation. Bags of one type only shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Five hundred bags of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, end opening, V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Five hundred bags of one type only, packaged as specified in 5.1, shall be baled by wrapping with two sheets of kraft paper, each sheet having a basis weight of not less than 60 pounds per ream in accordance with A-A-203. The wraps shall overlap widthwise and lengthwise a minimum of 4-inches and shall be secured with tape conforming to PPP-T-76. Alternatively, five hundred bags may be packed directly into a fiberboard shipping container conforming to style RSC, end opening, type CF, variety SW, or type SF, class domestic, grade 350 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636. When specified (see 6.2), the shipping container shall be V2s, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

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5.2.3 Commercial packing. The paper bags, preserved as specified in 5.1, shall be packed in accordance with ASTM-D-3951.

5.3 Palletization. When specified (see 6.2), paper bags, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type Ia (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C and D or film bonding F or G. Pallet patterns shall be in accordance with the appendix B of MIL-STD-147. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Waxed paper bags are intended for use in the delivery of 6-loaf sheets of field bread from the mobile or field bakery to the using units.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Shipping container required (see 5.2.2).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

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6.5 Subject term (key word) listing.

Baked goods
Bakery
Loaf
Mobile bakery
Subsistence packaging

6.6 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA

Review activities:

Army - MD, SM
Air Force - 99

Preparing activity:

Army - GL

(Project 8105-0328)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-43611B	1990 December 31

3. DOCUMENT TITLE
BAGS, PAPER, WAXED (FIELD BREAD)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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